Zara Data Dictionary

Overview

The dataset covers Zara's 35 stores in seven countries (United States, China, Germany, France, Spain, Portugal, United Kingdom) from **1 January 2023 to 18 March 2025**: about 3.9 million rows in 6 CSV files. Sales are recorded in each store's local currency at Zara's local price list.

The data is simulated for teaching. It does not represent Zara's or Inditex's actual stores, customers, figures or events. Like real retail data, it contains quality issues; finding and handling them is part of the work.

Table	Rows	One row is
stores	35	a store
employees	1,746	a store employee
customers	591,159	a customer profile
products	7,199	a product (one style in one colour)
discounts	513	one markdown phase for one sub-category, in the listed countries
transactions	3,340,935	one line of a sale or return invoice

How the tables link

- stores.Store ID → employees.Store ID and transactions.Store ID
- employees.Employee ID → transactions.Employee ID
- customers.Customer ID → transactions.Customer ID
- products.Product ID → transactions.Product ID
- discounts has no key column. It links to transactions by date (between Start and End), to products by Category and Sub Category, and to stores by Country (blank Country = all countries)

Stores, employees and customers

stores

Column	Description
Store ID	Primary key
Country	Store country. Some names are in the local language or script (中国, Deutschland, España)
City	Store city, in the local spelling (e.g. München, Lisboa, 上海)
Store Name	e.g. "Zara Fifth Avenue", "Zara Gran Vía"
Number of Employees	Headcount recorded for the store
ZIP Code	Postal code
Latitude, Longitude	Store location, for maps

employees

Column	Description
Employee ID	Primary key
Store ID	→ stores
Name	Employee name
Position	Store Manager, Department Manager, Visual Merchandiser, Cashier, Stock Associate, Sales Associate

customers

Column	Description
Customer ID	Primary key

Column	Description
Name, Email, Telephone	Contact details (fictional; emails use the reserved example.com domain, phone formats vary by country)
City, Country	Where the customer lives. This can differ from the store they shop in, for example tourists
Gender	F, M, or D (diverse)
Date Of Birth	Use it to derive age and age bands
Job Title	Occupation; blank for about a third of customers

Products, discounts and transactions

products

Column	Description
Product ID	Primary key
Category	Woman, Man, Kids
Sub Category	29 in total, e.g. Dresses, Blazers, Jeans, Knitwear, Shoes, Perfumes (Woman); Coats and Jackets, Overshirts (Man); Girls, Boys, Baby (Kids)
Description EN	Product name, e.g. "ZW COLLECTION MIDI DRESS", "WOOL BLEND COAT"
Color	Colour of this product (each colour is a separate product)
Sizes	Sizes it comes in, separated by |: letters (XS|S|M|L|XL), EU jeans and shoe sizes, kids' ages (4-5Y|6-7Y…), baby months (0-3M…), ONE SIZE or 100 ML
Production Cost	Cost to make one unit, in EUR
Launch Date	First day the product could be sold

discounts

Column	Description
Start, End	First and last day of the markdown phase
Discont	Markdown rate, e.g. 0.4 = 40% (the column name is misspelled in the file)
Description	Sale season and phase, e.g. "40% off during our Winter Sale 2023-24 - phase 2"
Category, Sub Category	Products the markdown applies to
Country	Countries where the phase ran; blank = all seven countries

transactions

Column	Description
Invoice ID	Invoice number. INV- for sales, RET- for returns, then country code, store number in that country and a sequence (e.g. INV-US-001-03158761)
Line	Line number within the invoice
Customer ID	→ customers
Product ID	→ products
Size, Color	Size and colour bought
Unit Price	Price of one unit before discount, in the store's currency
Quantity	Units on the line
Date	Date and time of the transaction
Discount	Markdown applied to the line, e.g. 0.5 = 50%; 0 = full price
Line Total	Unit Price × Quantity × (1 − Discount). Negative on return lines

Column	Description
Store ID	→ stores
Employee ID	→ employees (who served or processed the transaction)
Currency, Currency Symbol	USD ($), EUR (€), GBP (£) or CNY (¥)
SKU	Stock-keeping unit: category and sub-category code, product ID, size and colour code (e.g. WODR1234-M-BLA)
Transaction Type	Sale or Return
Payment Method	Cash, Credit Card or Mobile Payment
Invoice Total	Total of the whole invoice, repeated on every line of that invoice

Business rules

1. **Invoices and lines.** One invoice has one or more lines. Count invoices with a distinct count of Invoice ID; sum money with Line Total, never Invoice Total (it repeats on every line).
2. **Returns.** A return is a separate RET- invoice, made within 30 days of the purchase, usually at the same store. It copies the original line's product, size, price and discount, with negative totals. Net sales = sales + returns.
3. **Currencies.** Every sales amount is in the store's local currency; Production Cost is in EUR. To compare stores or countries, convert to US dollars with these fixed average rates: 1 EUR = 1.08 USD, 1 GBP = 1.27 USD, 1 CNY = 0.14 USD.
4. **Markdowns.** Zara marks prices down only during its sale seasons: the Winter Sale (from late December) and the Summer Sale (from late June), each in three phases of deepening discounts, plus Black Friday 2024 in the US and UK. Only older products go on markdown; new arrivals stay at full price.
5. **Date range.** The data runs from 1 January 2023 to 18 March 2025, so 2025 is a partial year. Compare it only with the same dates in 2024.
6. **Gross margin** = net sales − Production Cost × Quantity, with both converted to the same currency.